

Serveur Mail Complet avec Postfix

Domaine	afj.local
Hostname	mail.afj.local
Ip serveur	192.168.1.54
Os server	ubuntu server 24.04

Préparation du système

```
dbsrv@dbsrv:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for dbsrv:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
16 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-headers-6.8.0-117 linux-headers-6.8.0-117-generic linux-image-6.8.0-117-generic
  linux-modules-6.8.0-117-generic linux-modules-extra-6.8.0-117-generic linux-tools-6.8.0-117
  linux-tools-6.8.0-117-generic
The following packages will be upgraded:
  distro-info-data iproute2 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1
  linux-firmware linux-generic linux-headers-generic linux-image-generic linux-libc-dev linux-tools-common
  open-vm-tools thermald ubuntu-pro-client ubuntu-pro-client-l10n
16 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 835 MB of archives.
After this operation, 309 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iproute2 amd64 6.1.0-1ubuntu6.3 [1,120 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 open-vm-tools amd64 2:13.0.0-2ubuntu0.24.04.1 [723 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 distro-info-data all 0.60ubuntu0.6 [7,036 B]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ubuntu-pro-client-l10n amd64 37.2ubuntu-24.04 [19.8 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ubuntu-pro-client amd64 37.2ubuntu-24.04 [259 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomenc amd64 1.17.6-1ubuntu4.3 [14.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomdec amd64 1.17.6-1ubuntu4.3 [10.6 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif1 amd64 1.17.6-1ubuntu4.3 [276 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-libde265 amd64 1.17.6-1ubuntu4.3 [8,156 B]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmware amd64 20240318.git3b128b60-0ubuntu2.27 [641 MB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-6.8.0-117-generic amd64 6.8.0-117.117 [39.1 MB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-6.8.0-117-generic amd64 6.8.0-117.117 [14.8 MB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-extra-6.8.0-117-generic amd64 6.8.0-117.117 [113 MB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-generic amd64 6.8.0-117.117 [11,698 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-generic amd64 6.8.0-117.117 [11.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.8.0-117 all 6.8.0-117.117 [13.6 MB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.8.0-117-generic amd64 6.8.0-117.117 [3,755 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-generic amd64 6.8.0-117.117 [11.1 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-libc-dev amd64 6.8.0-117.117 [1,495 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-117.117 [335 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-117 amd64 6.8.0-117.117 [5,000 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-117-generic amd64 6.8.0-117.117 [1,814 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 thermald amd64 2.5.6-2ubuntu0.24.04.5 [214 kB]
Fetched 835 MB in 47s (17.7 MB/s)
Preconfiguring packages ...
(Reading database ... 126029 files and directories currently installed.)
Preparing to unpack .../00-iproute2_6.1.0-1ubuntu6.3_amd64.deb ...
Unpacking iproute2 (6.1.0-1ubuntu6.3) over (6.1.0-1ubuntu6.2) ...
Preparing to unpack .../01-open-vm-tools_23a13.0.0-2ubuntu0.24.04.1_amd64.deb ...
Unpacking open-vm-tools (2:13.0.0-2ubuntu0.24.04.1) over (2:12.5.0-1ubuntu0.24.04.2) ...
```

```
dbsrv@dbsrv:~  
Preconfiguring packages ...  
(Reading database ... 126029 files and directories currently installed.)  
Preparing to unpack .../00-iproute2_6.1.0-1ubuntu6.3_amd64.deb ...  
Unpacking iproute2 (6.1.0-1ubuntu6.3) over (6.1.0-1ubuntu6.2) ...  
Preparing to unpack .../01-open-vm-tools_2k3a13.0.0-2ubuntu0.24.04.1_amd64.deb ...  
Unpacking open-vm-tools (2:13.0.0-2-ubuntu0.24.04.1) over (2:12.5.0-1-ubuntu0.24.04.2) ...  
Preparing to unpack .../02-distro-info-data_0.60ubuntu0.6_all.deb ...  
Unpacking distro-info-data (0.60ubuntu0.6) over (0.60ubuntu0.5) ...  
Preparing to unpack .../03-ubuntu-pro-client-110n_37.2ubuntu-24.04_amd64.deb ...  
Unpacking ubuntu-pro-client-110n (37.2ubuntu-24.04) over (37.1ubuntu0-24.04) ...  
Preparing to unpack .../04-ubuntu-pro-client-37.2ubuntu-24.04_amd64.deb ...  
Unpacking ubuntu-pro-client (37.2ubuntu-24.04) over (37.1ubuntu0-24.04) ...  
Preparing to unpack .../05-libheif-plugin-aomdec_1.17.6-1ubuntu4.3_amd64.deb ...  
Unpacking libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.3) over (1.17.6-1ubuntu4.2) ...  
Preparing to unpack .../06-libheif-plugin-aomdec_1.17.6-1ubuntu4.3_amd64.deb ...  
Unpacking libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.3) over (1.17.6-1ubuntu4.2) ...  
Preparing to unpack .../07-libheif1_1.17.6-1ubuntu4.3_amd64.deb ...  
Unpacking libheif1:amd64 (1.17.6-1ubuntu4.3) over (1.17.6-1ubuntu4.2) ...  
Preparing to unpack .../08-libheif-plugin-libde265_1.17.6-1ubuntu4.3_amd64.deb ...  
Unpacking libheif-plugin-libde265:amd64 (1.17.6-1ubuntu4.3) over (1.17.6-1ubuntu4.2) ...  
Preparing to unpack .../09-linux-firmware_20240318.git3b128b60-0ubuntu2.27_amd64.deb ...  
Unpacking linux-firmware (20240318.git3b128b60-0ubuntu2.27) over (20240318.git3b128b60-0ubuntu2.26) ...  
Selecting previously unselected package linux-modules-6.8.0-117-generic.  
Preparing to unpack .../10-linux-modules-6.8.0-117-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-modules-6.8.0-117-generic (6.8.0-117.117) ...  
Selecting previously unselected package linux-image-6.8.0-117-generic.  
Preparing to unpack .../11-linux-image-6.8.0-117-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-image-6.8.0-117-generic (6.8.0-117.117) ...  
Selecting previously unselected package linux-modules-extra-6.8.0-117-generic.  
Preparing to unpack .../12-linux-modules-extra-6.8.0-117-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-modules-extra-6.8.0-117-generic (6.8.0-117.117) ...  
Preparing to unpack .../13-linux-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-generic (6.8.0-117.117) over (6.8.0-111.111) ...  
Preparing to unpack .../14-linux-image-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-image-generic (6.8.0-117.117) over (6.8.0-111.111) ...  
Selecting previously unselected package linux-headers-6.8.0-117.  
Preparing to unpack .../15-linux-headers-6.8.0-117_6.8.0-117.117_all.deb ...  
Unpacking linux-headers-6.8.0-117 (6.8.0-117.117) ...  
Selecting previously unselected package linux-headers-6.8.0-117-generic.  
Preparing to unpack .../16-linux-headers-6.8.0-117-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-headers-6.8.0-117-generic (6.8.0-117.117) ...  
Preparing to unpack .../17-linux-headers-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-headers-generic (6.8.0-117.117) over (6.8.0-111.111) ...  
Preparing to unpack .../18-linux-libc-dev_6.8.0-117.117_amd64.deb ...  
Unpacking linux-libc-dev:amd64 (6.8.0-117.117) over (6.8.0-111.111) ...  
Preparing to unpack .../19-linux-tools-common_6.8.0-117.117_all.deb ...  
Unpacking linux-tools-common (6.8.0-117.117) over (6.8.0-111.111) ...  
Selecting previously unselected package linux-tools-6.8.0-117.  
Preparing to unpack .../20-linux-tools-6.8.0-117_6.8.0-117.117_amd64.deb ...  
Unpacking linux-tools-6.8.0-117 (6.8.0-117.117) ...  
Selecting previously unselected package linux-tools-6.8.0-117-generic.  
Preparing to unpack .../21-linux-tools-6.8.0-117-generic_6.8.0-117.117_amd64.deb ...  
Unpacking linux-tools-6.8.0-117-generic (6.8.0-117.117) ...  
Preparing to unpack .../22-thermald_2.5.6-2ubuntu0.24.04.5_amd64.deb ...  
Unpacking thermald (2.5.6-2ubuntu0.24.04.5) over (2.5.6-2ubuntu0.24.04.3) ...
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```
Installing new version of config file /etc/apparmor.d/ubuntu_pro_esa_cache ...  
Setting up linux-modules-6.8.0-117-generic (6.8.0-117.117) ...  
Setting up ubuntu-pro-client-110n (37.2ubuntu-24.04) ...  
Setting up linux-tools-6.8.0-117 (6.8.0-117.117) ...  
Setting up linux-image-6.8.0-117-generic (6.8.0-117.117) ...  
I: /boot/vmlinuz.old is now a symlink to vmlinuz-6.8.0-111-generic  
I: /boot/initrd.img.old is now a symlink to initrd.img-6.8.0-111-generic  
I: /boot/vmlinuz is now a symlink to vmlinuz-6.8.0-117-generic  
I: /boot/initrd.img is now a symlink to initrd.img-6.8.0-117-generic  
Setting up linux-modules-extra-6.8.0-117-generic (6.8.0-117.117) ...  
Setting up linux-tools-6.8.0-117-generic (6.8.0-117.117) ...  
Setting up linux-headers-generic (6.8.0-117.117) ...  
Setting up linux-image-generic (6.8.0-117.117) ...  
Setting up linux-generic (6.8.0-117.117) ...  
Setting up libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.3) ...  
Setting up libheif1:amd64 (1.17.6-1ubuntu4.3) ...  
Setting up libheif-plugin-libde265:amd64 (1.17.6-1ubuntu4.3) ...  
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.3) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for dbus (1.14.10-4ubuntu1) ...  
Processing triggers for initramfs-tools (0.142ubuntu25.8) ...  
update-initramfs: Generating /boot/initrd.img-6.8.0-111-generic  
Processing triggers for linux-image-6.8.0-117-generic (6.8.0-117.117) ...  
/etc/kernel/postinst.d/initramfs-tools:  
update-initramfs: Generating /boot/initrd.img-6.8.0-117-generic  
/etc/kernel/postinst.d/22-update-grub:  
Sourcing file '/etc/default/grub'  
Generating grub configuration file ...  
Found linux image: /boot/vmlinuz-6.8.0-117-generic  
Found initrd image: /boot/initrd.img-6.8.0-117-generic  
Found linux image: /boot/vmlinuz-6.8.0-111-generic  
Found initrd image: /boot/initrd.img-6.8.0-111-generic  
Found linux image: /boot/vmlinuz-6.8.0-110-generic  
Found initrd image: /boot/initrd.img-6.8.0-110-generic  
Warning: os-prober will not be executed to detect other bootable partitions.  
Systems on them will not be added to the GRUB boot configuration.  
Check GRUB_DISABLE_OS_PROBER documentation entry.  
Adding boot menu entry for UEFI Firmware Settings ...  
done  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
6.8.0-110-generic  
Diagnostics:  
The currently running kernel version is not the expected kernel version 6.8.0-117-generic.  
  
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.  
  
Restarting services...  
  
Service restarts being deferred:
```

```

Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-119-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
dbsrv @ session #1: login[1270]
dbsrv @ user manager service: systemd[1418]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$

```

Installation de Apache2 et Nginx

```

dbsrv@dbsrv:~$ sudo apt update && sudo apt upgrade -y \
&& sudo apt install -y apache2 nginx
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-111 linux-headers-6.8.0-111-generic linux-image-6.8.0-111-generic linux-modules-6.8.0-111-generic linux-modules-extra-6.8.0-111-generic linux-tools-6.8.0-111 linux-tools-6.8.0-111-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-111 linux-headers-6.8.0-111-generic linux-image-6.8.0-111-generic linux-modules-6.8.0-111-generic linux-modules-extra-6.8.0-111-generic linux-tools-6.8.0-111 linux-tools-6.8.0-111-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 nginx-common ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser fcgiwrap nginx-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 nginx nginx-common ssl-cert
0 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 2,658 kB of archives.
After this operation, 9,716 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [188 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.0-3build2 [166 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.12 [1,336 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.12 [163 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.12 [98.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.12 [90.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.8 [43.9 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.8 [522 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2,658 kB in 1s (3,096 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 164860 files and directories currently installed.)
Preparing to unpack .../00-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../01-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...

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Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../01-libaprutil1t64_1.6.3-1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../02-libaprutil1-dbd-sqlite3_1.6.3-1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../03-libaprutil1-ldap_1.6.3-1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1ubuntu7) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../04-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../05-apache2-bin_2.4.58-1ubuntu8.12_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu8.12) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../06-apache2-data_2.4.58-1ubuntu8.12_all.deb ...
Unpacking apache2-data (2.4.58-1ubuntu8.12) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../07-apache2-utils_2.4.58-1ubuntu8.12_amd64.deb ...
Unpacking apache2-utils (2.4.58-1ubuntu8.12) ...
Selecting previously unselected package apache2.
Preparing to unpack .../08-apache2_2.4.58-1ubuntu8.12_amd64.deb ...
Unpacking apache2 (2.4.58-1ubuntu8.12) ...
Selecting previously unselected package nginx-common.
Preparing to unpack .../09-nginx-common_1.24.0-2ubuntu7.8_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.8) ...
Selecting previously unselected package nginx.
Preparing to unpack .../10-nginx_1.24.0-2ubuntu7.8_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.8) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../11-ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Setting up nginx-common (1.24.0-2ubuntu7.8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.service.
Setting up libaprutil1t64:amd64 (1.6.3-1ubuntu7) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1ubuntu7) ...
Setting up apache2-data (2.4.58-1ubuntu8.12) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-1ubuntu7) ...
Setting up nginx (1.24.0-2ubuntu7.8) ...
* Upgrading binary nginx
Setting up libaprutil1-ldap:amd64 (1.6.3-1ubuntu7) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1ubuntu7) ...
Setting up apache2-utils (2.4.58-1ubuntu8.12) ...
Setting up apache2-bin (2.4.58-1ubuntu8.12) ...
Setting up apache2 (2.4.58-1ubuntu8.12) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
```

[OK]

```

Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-118-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
dbsrv @ session #1: login[1270]
dbsrv @ user manager service: systemd[1418]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$
```

Installation de Postfix

```
dbsrv@dbsrv:~$ sudo apt-get install postfix
[sudo] password for dbsrv:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-111 linux-headers-6.8.0-111-generic linux-image-6.8.0-111-generic linux-modules-6.8.0-111-generic linux-modules-extra-6.8.0-111-generic linux-tools-6.8.0-111 linux-tools-6.8.0-111-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libsasl2
Suggested packages:
  mail-reader postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql postfix-pcre postfix-pgsql postfix-sqlite procmail sasl2-bin | dovecot-common
The following NEW packages will be installed:
  libsasl2 postfix
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,296 kB of archives.
After this operation, 4,321 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libsasl2 amd64 1:3.0-3build3 [41.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 postfix amd64 3.8.6-1ubuntu0.1 [1,255 kB]
Fetched 1,296 kB in 1s (1,426 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libsasl2:amd64.
(Reading database ... 165629 files and directories currently installed.)
Preparing to unpack .../libsasl2_1:3.0-3build3_amd64.deb ...
Unpacking libsasl2:amd64 (1:3.0-3build3) ...
Selecting previously unselected package postfix.
Preparing to unpack .../postfix_3.8.6-1ubuntu0.1_amd64.deb ...
Unpacking postfix (3.8.6-1ubuntu0.1) ...
Setting up libsasl2:amd64 (1:3.0-3build3) ...
Setting up postfix (3.8.6-1ubuntu0.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'postfix' (GID 111) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user 'postfix' (UID 110) ...
info: Adding new user 'postfix' (UID 110) with group 'postfix' ...
info: Not creating home directory '/var/spool/postfix'.
Creating /etc/postfix/dynamicmaps.cf
info: Selecting GID from range 100 to 999 ...
info: Adding group 'postdrop' (GID 112) ...

setting myhostname: dbsrv.lan
setting alias maps
setting alias database
changing /etc/mailname to mail.afj.local
setting myorigin
setting destinations: $myhostname, mail.afj.local, dbsrv, localhost.localdomain, localhost
setting relayhost:
setting mynetworks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
setting mailbox_size_limit: 0
setting recipient_delimiter: +
setting inet_interfaces: all
setting inet_protocols: all
/etc/aliases does not exist, creating it.
WARNING: /etc/aliases exists, but does not have a root alias.
```

```
Postfix (main.cf) is now set up with a default configuration. If you need to
make changes, edit /etc/postfix/main.cf (and others) as needed. To view
Postfix configuration values, see postconf(1).

After modifying main.cf, be sure to run 'systemctl reload postfix'.

Running newaliases
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
Processing triggers for rsyslog (8.2312.0-3ubuntu9.2) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-110-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

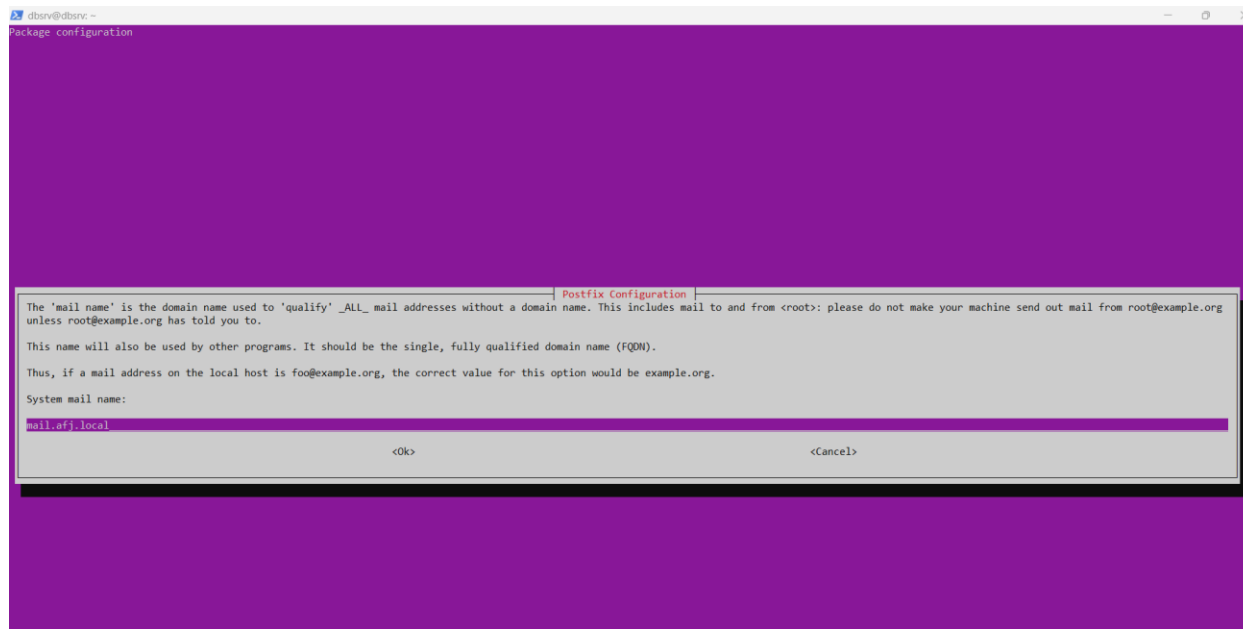
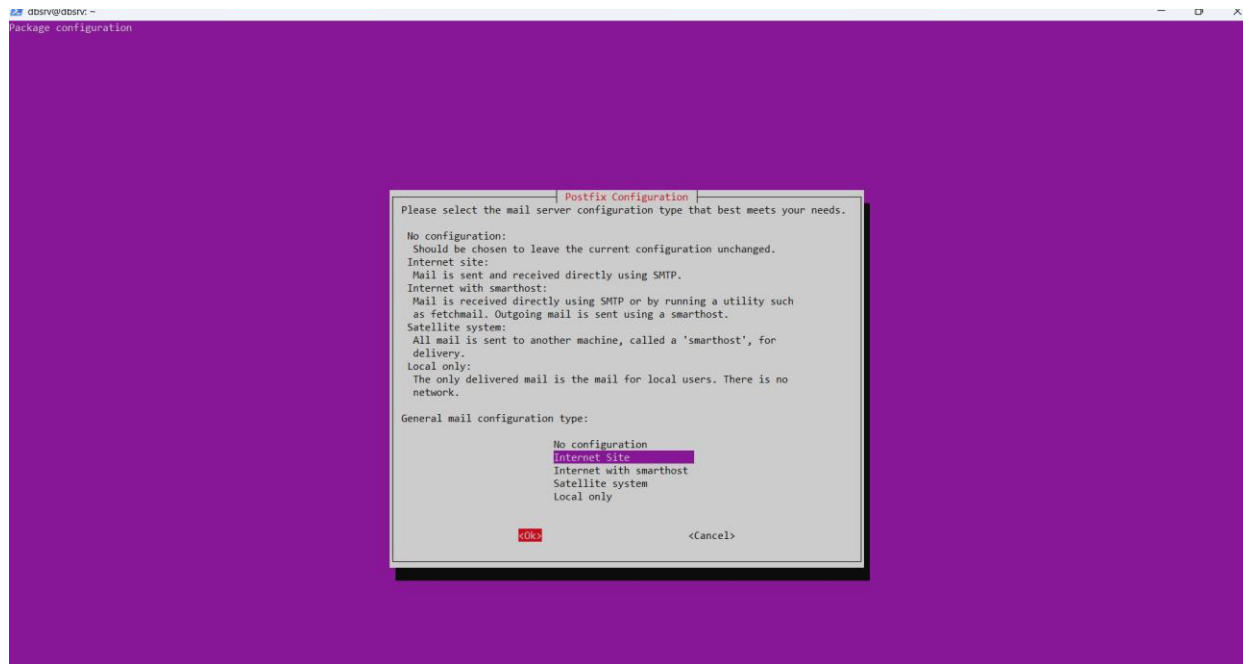
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
dbsrv @ session #1: login[1270]
dbsrv @ user manager service: systemd[1418]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$
```



Configuration



```
dbsrv@dbsrv: ~  
GNU nano 7.2 /etc/postfix/main.cf  
# See /usr/share/postfix/main.cf.dist for a commented, more complete version  
  
# Debian specific: Specifying a file name will cause the first  
# line of that file to be used as the name. The Debian default  
# is /etc/mailname.  
#myorigin = /etc/mailname  
  
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)  
biff = no  
  
# appending .domain is the MUA's job.  
append_dot_mydomain = no  
  
# Uncomment the next line to generate "delayed mail" warnings  
#delay_warning_time = 4h  
  
readme_directory = no  
  
# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on  
# fresh installs.  
compatibility_level = 3.6  
  
# TLS parameters  
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem  
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key  
smtpd_tls_security_level=may  
  
smtp_tls_CApath=/etc/ssl/certs  
smtp_tls_security_level=may  
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache  
  
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination  
myhostname = dbsrv.lan  
alias_maps = hash:/etc/aliases  
alias_database = hash:/etc/aliases  
myorigin = /etc/mailname  
mydestination = $myhostname, mail.afj.local, dbsrv, localhost.localdomain, localhost  
relayhost =  
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128  
mailbox_size_limit = 0  
recipient_delimiter = +  
inet_interfaces = all  
inet_protocols = all  
  
myhostname = mail.afj.local  
mydomain = afj.local  
myorigin = $mydomain  
[ Read 55 lines ]  
^G Help      ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo  
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```


Redémarrage de Postfix

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo systemctl restart postfix  
dbsrv@dbsrv:~$
```

Vérification du port de Postfix

```
dbsrv@dbsrv:~$ sudo ss -tlnp | grep master  
LISTEN 0      100          0.0.0.0:25      0.0.0.0:*      users:((("master",pid=44687,fd=13))  
  
LISTEN 0      100          [::]:25        [::]:*         users:((("master",pid=44687,fd=14))  
  
dbsrv@dbsrv:~$
```

Installation de Dovecot (IMAP / POP3)

```
dbsrv@dbsrv:~  
dbsrv@dbsrv:~$ sudo apt install dovecot-imapd dovecot-pop3d -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  linux-headers-6.8.0-111 linux-headers-6.8.0-111-generic linux-image-6.8.0-111-generic  
  linux-modules-6.8.0-111-generic linux-modules-extra-6.8.0-111-generic linux-tools-6.8.0-111  
  linux-tools-6.8.0-111-generic  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  dovecot-core libexttextcat-2.0-0 libexttextcat-data  
Suggested packages:  
  dovecot-gssapi dovecot-ldap dovecot-lmtpd dovecot-managesieved dovecot-mysql dovecot-pgsql  
  dovecot-sieve dovecot-solr dovecot-sqlite dovecot-submissiond ntp  
The following NEW packages will be installed:  
  dovecot-core dovecot-imapd dovecot-pop3d libexttextcat-2.0-0 libexttextcat-data  
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.  
Need to get 3,786 kB of archives.  
After this operation, 12.1 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libexttextcat-data all 3.4.7-1build1 [189 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libexttextcat-2.0-0 amd64 3.4.7-1build1 [13.3 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-core amd64 1:2.3.21+dfsg1-2ubuntu6.4 [3,351 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-imapd amd64 1:2.3.21+dfsg1-2ubuntu6.4 [191 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-pop3d amd64 1:2.3.21+dfsg1-2ubuntu6.4 [37.3 kB]  
Fetched 3,786 kB in 1s (3,604 kB/s)  
Selecting previously unselected package libexttextcat-data.  
(Reading database ... 155837 files and directories currently installed.)  
Preparing to unpack .../libexttextcat-data_3.4.7-1build1_all.deb ...  
Unpacking libexttextcat-data (3.4.7-1build1) ...  
Selecting previously unselected package libexttextcat-2.0-0:amd64.  
Preparing to unpack .../libexttextcat-2.0-0_3.4.7-1build1_amd64.deb ...  
Unpacking libexttextcat-2.0-0:amd64 (3.4.7-1build1) ...  
Selecting previously unselected package dovecot-core.  
Preparing to unpack .../dovecot-core_1%3a2.3.21+dfsg1-2ubuntu6.4_amd64.deb ...  
Unpacking dovecot-core (1:2.3.21+dfsg1-2ubuntu6.4) ...  
Selecting previously unselected package dovecot-imapd.  
Preparing to unpack .../dovecot-imapd_1%3a2.3.21+dfsg1-2ubuntu6.4_amd64.deb ...  
Unpacking dovecot-imapd (1:2.3.21+dfsg1-2ubuntu6.4) ...  
Selecting previously unselected package dovecot-pop3d.  
Preparing to unpack .../dovecot-pop3d_1%3a2.3.21+dfsg1-2ubuntu6.4_amd64.deb ...  
Unpacking dovecot-pop3d (1:2.3.21+dfsg1-2ubuntu6.4) ...  
Setting up libexttextcat-data (3.4.7-1build1) ...  
Setting up libexttextcat-2.0-0:amd64 (3.4.7-1build1) ...  
Setting up dovecot-core (1:2.3.21+dfsg1-2ubuntu6.4) ...  
  
Creating config file /etc/dovecot/dovecot.conf with new version  
  
Creating config file /etc/dovecot/dovecot-dict-auth.conf.ext with new version  
  
Creating config file /etc/dovecot/dovecot-dict-sql.conf.ext with new version  
  
Creating config file /etc/dovecot/dovecot-sql.conf.ext with new version  
  
Creating config file /etc/dovecot/conf.d/10-auth.conf with new version
```



```

dbsrv@dbsrv: ~
Creating config file /etc/dovecot/dovecot-dict-sql.conf.ext with new version
Creating config file /etc/dovecot/dovecot-sql.conf.ext with new version
Creating config file /etc/dovecot/conf.d/10-auth.conf with new version
Creating config file /etc/dovecot/conf.d/10-director.conf with new version
Creating config file /etc/dovecot/conf.d/10-logging.conf with new version
Creating config file /etc/dovecot/conf.d/10-mail.conf with new version
Creating config file /etc/dovecot/conf.d/10-master.conf with new version
Creating config file /etc/dovecot/conf.d/10-ssl.conf with new version
Creating config file /etc/dovecot/conf.d/10-tcpwrapper.conf with new version
Creating config file /etc/dovecot/conf.d/15-lda.conf with new version
Creating config file /etc/dovecot/conf.d/15-mailboxes.conf with new version
Creating config file /etc/dovecot/conf.d/90-acl.conf with new version
Creating config file /etc/dovecot/conf.d/90-plugin.conf with new version
Creating config file /etc/dovecot/conf.d/90-quota.conf with new version
Creating config file /etc/dovecot/conf.d/auth-checkpassword.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-deny.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-dict.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-master.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-passwdfile.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-sql.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-static.conf.ext with new version
Creating config file /etc/dovecot/conf.d/auth-system.conf.ext with new version
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
dovecot.socket is a disabled or a static unit, not starting it.
Setting up dovecot-imapd (1:2.3.21+dfsg1-2ubuntu6.4) ...
Creating config file /etc/dovecot/conf.d/20-imap.conf with new version
Setting up dovecot-pop3d (1:2.3.21+dfsg1-2ubuntu6.4) ...
Creating config file /etc/dovecot/conf.d/20-pop3.conf with new version
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...

```

```

dbsrv@dbsrv: ~
Creating config file /etc/dovecot/conf.d/20-imap.conf with new version
Setting up dovecot-pop3d (1:2.3.21+dfsg1-2ubuntu6.4) ...

Creating config file /etc/dovecot/conf.d/20-pop3.conf with new version
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for dovecot-core (1:2.3.21+dfsg1-2ubuntu6.4) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-110-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
dbsrv @ session #1: login[1270]
dbsrv @ user manager service: systemd[1418]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$

```

```

dbsrv@dbsrv:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-18 02:22:56 UTC; 5min ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
  Main PID: 48138 (dovecot)
    Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 2205)
  Memory: 3.4M (peak: 3.7M)
     CPU: 26ms
  CGroup: /system.slice/dovecot.service
          └─48138 /usr/sbin/dovecot -F
            └─48139 dovecot/anvil
              └─48140 dovecot/log
                └─48141 dovecot/config

May 18 02:22:56 dbsrv systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
May 18 02:22:56 dbsrv dovecot[48138]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3 (core dumps disabled)
May 18 02:22:56 dbsrv systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
dbsrv@dbsrv:~$

```

Configuration

```
GNU nano 7.2 /etc/dovecot/conf.d/10-mail.conf
# Maximum mbox file size until it's rotated.
#mbox_rotate_size = 10M

# Maximum mbox file age until it's rotated. Typically in days. Day begins
# from midnight, so 1d = today, 2d = yesterday, etc. 0 = check disabled.
#mbox_rotate_interval = 0

# When creating new mbox files, immediately preallocate their size to
# mbox_rotate_size. This setting currently works only in linux with some
# filesystems (ext4, xfs).
#mbox_preallocate_space = no

##
## Mail attachments
##

# mbox and mbox support saving mail attachments to external files, which
# also allows single instance storage for them. Other backends don't support
# this for now.

# Directory root where to store mail attachments. Disabled, if empty.
#mail_attachment_dir =

# Attachments smaller than this aren't saved externally. It's also possible to
# write a plugin to disable saving specific attachments externally.
#mail_attachment_min_size = 128k

# Filesystem backend to use for saving attachments:
# posix : No SIS done by Dovecot (but this might help FS's own deduplication)
# sis-queue : SIS with immediate byte-by-byte comparison during saving
# sis-posix : SIS with delayed comparison and deduplication
#mail_attachment_fs = sis-posix

# Hash format to use in attachment filenames. You can add any text and
# variables: %(md4), %(md5), %(sha1), %(sha256), %(sha512), %(size).
# Variables can be truncated, e.g. %(sha256:80) returns only first 80 bits
#mail_attachment_hash = %(sha1)

# Settings to control adding $MboxAttachment or $MboxBoltAttachment keywords.
# By default, all MIME parts with Content-Disposition=attachment, or inlines
# with filename parameter are considered attachments.
# add-flags - Add the keywords when saving new mails or when fetching can
# do it efficiently.
# content-type-type or ltype - Include/exclude content type. Excluding will
# never consider the matched MIME part as attachment. Including will only
# negate an exclusion (e.g. content-type=!foo/* content-type=foo/bar).
# exclude-inlined - Exclude any Content-Disposition=inline MIME part.
#mail_attachment_detection_options =
mail_location = maildir:~/Maildir
```

Figure 1-mail_driver et mail_path ont été remplacés par mail_location = maildir:~/Maildir

Protocoles

```
GNU nano 7.2 /etc/dovecot/dovecot.conf
# proxying. This isn't necessary normally, but may be useful if the destination
# IP is e.g. a load balancer's IP.
#auth_proxy_self =

# Show more verbose process titles (in ps). Currently shows user name and
# IP address. Useful for seeing who are actually using the IMAP processes
# (eg. shared mailboxes or if same uid is used for multiple accounts).
#verbose_proctitle = no

# Should all processes be killed when Dovecot master process shuts down.
# Setting this to "no" means that Dovecot can be upgraded without
# forcing existing client connections to close (although that could also be
# a problem if the upgrade is e.g. because of a security fix).
#shutdown_clients = yes

# If non-zero, run mail commands via this many connections to doveadm server,
# instead of running them directly in the same process.
#doveadm_worker_count = 0
# UNIX socket or host:port used for connecting to doveadm server.
#doveadm_socket_path = doveadm-server

# Space separated list of environment variables that are preserved on Dovecot
# startup and passed down to all of its child processes. You can also give
# key=value pairs to always set specific settings.
#import_environment = TZ

##
## Dictionary server settings
##

# Dictionary can be used to store key-value lists. This is used by several
# plugins. The dictionary can be accessed either directly or through a
# dictionary server. The following dict block maps dictionary names to URIs
# when the server is used. These can then be referenced using URIs in format
# "proxy::<name>".

dict {
  #quota = mysql:/etc/dovecot/dovecot-dict-sql.conf.ext
}

# Most of the actual configuration gets included below. The filenames are
# first sorted by their ASCII value and parsed in that order. The 00-prefixes
# in filenames are intended to make it easier to understand the ordering.
!include conf.d/*.conf

# A config file can also be tried to be included without giving an error if
# it's not found:
!include_try local.conf
protocols = imap pop3
```

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket Previous Back Prev Word
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was Next Forward Next Word

Redémarrage

```
dbsrv@dbsrv:~$ sudo systemctl restart dovecot
dbsrv@dbsrv:~$
```

Ports de Dovecot

```
dbsrv@dbsrv:~$ sudo ss -tlnp | grep dovecot
LISTEN 0      100          0.0.0.0:995      0.0.0.0:*      users:((("dovecot",pid=48857,fd=24))
LISTEN 0      100          0.0.0.0:993      0.0.0.0:*      users:((("dovecot",pid=48857,fd=41))
LISTEN 0      100          0.0.0.0:143      0.0.0.0:*      users:((("dovecot",pid=48857,fd=39))
LISTEN 0      100          0.0.0.0:110      0.0.0.0:*      users:((("dovecot",pid=48857,fd=22))
LISTEN 0      100          [::]:995         [::]:*         users:((("dovecot",pid=48857,fd=25))
LISTEN 0      100          [::]:993         [::]:*         users:((("dovecot",pid=48857,fd=42))
LISTEN 0      100          [::]:143         [::]:*         users:((("dovecot",pid=48857,fd=40))
LISTEN 0      100          [::]:110         [::]:*         users:((("dovecot",pid=48857,fd=23))
dbsrv@dbsrv:~$
```

Dovecot écoute sur plusieurs ports avec par exemple :

- ❖ **110/tcp** → POP3 (non chiffré)
- ❖ **995/tcp** → POP3S (POP3 sécurisé via SSL/TLS)
- ❖ **993/tcp** → IMAPS (IMAP sécurisé via SSL/TLS)

Création des comptes utilisateurs

Création de 8 utilisateurs : direction, secretariat, mediatheque, comptabilite, coordination, professeur, etudiants, user01

```
dbsrv@dbsrv:~$ sudo adduser direction
[sudo] password for dbsrv:
info: Adding user `direction' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `direction' (1001) ...
info: Adding new user `direction' (1001) with group `direction (1001)' ...
info: Creating home directory `/home/direction' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for direction
Enter the new value, or press ENTER for the default
  Full Name []: Elisabeth Saint-Val
    Room Number []: 1
    Work Phone []: 42184534
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `direction' to supplemental / extra groups `users' ...
info: Adding user `direction' to group `users' ...
dbsrv@dbsrv:~$
```

```
dbsrv@dbsrv:~$ sudo adduser secretariat
info: Adding user `secretariat' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `secretariat' (1002) ...
info: Adding new user `secretariat' (1002) with group `secretariat (1002)' ...
info: Creating home directory `/home/secretariat' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for secretariat
Enter the new value, or press ENTER for the default
    Full Name []: Mercile Dorval
    Room Number []: 2
    Work Phone []: 47342181
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `secretariat' to supplemental / extra groups `users' ...
info: Adding user `secretariat' to group `users' ...
dbsrv@dbsrv:~$ sudo adduser mediatheque
info: Adding user `mediatheque' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `mediatheque' (1003) ...
info: Adding new user `mediatheque' (1003) with group `mediatheque (1003)' ...
info: Creating home directory `/home/mediatheque' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mediatheque
Enter the new value, or press ENTER for the default
    Full Name []: Lancia Pierre
    Room Number []: 3
    Work Phone []: 34328915
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `mediatheque' to supplemental / extra groups `users' ...
info: Adding user `mediatheque' to group `users' ...
dbsrv@dbsrv:~$
```

dbsrv@dbsrv: ~

```
dbsrv@dbsrv:~$ sudo adduser comptabilite
info: Adding user `comptabilite' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `comptabilite' (1004) ...
info: Adding new user `comptabilite' (1004) with group `comptabilite (1004)' ...
info: Creating home directory `/home/comptabilite' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for comptabilite
Enter the new value, or press ENTER for the default
    Full Name []: Peterson Paul
    Room Number []: 3
    Work Phone []: 44153231
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `comptabilite' to supplemental / extra groups `users' ...
info: Adding user `comptabilite' to group `users' ...
dbsrv@dbsrv:~$ sudo adduser coordination
info: Adding user `coordination' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `coordination' (1005) ...
info: Adding new user `coordination' (1005) with group `coordination (1005)' ...
info: Creating home directory `/home/coordination' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for coordination
Enter the new value, or press ENTER for the default
    Full Name []: Mecene Raymond
    Room Number []: 4
    Work Phone []: 37074415
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `coordination' to supplemental / extra groups `users' ...
info: Adding user `coordination' to group `users' ...
```

```
dbsrv@dbsrv:~$ sudo adduser professeur
info: Adding user `professeur' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `professeur' (1006) ...
info: Adding new user `professeur' (1006) with group `professeur (1006)' ...
info: Creating home directory `/home/professeur' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for professeur
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `professeur' to supplemental / extra groups `users' ...
info: Adding user `professeur' to group `users' ...
dbsrv@dbsrv:~$ sudo adduser etudiants
info: Adding user `etudiants' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `etudiants' (1007) ...
info: Adding new user `etudiants' (1007) with group `etudiants (1007)' ...
info: Creating home directory `/home/etudiants' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for etudiants
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `etudiants' to supplemental / extra groups `users' ...
info: Adding user `etudiants' to group `users' ...
```

```
dbsrv@dbsrv:~$ sudo adduser user01
info: Adding user `user01' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user01' (1008) ...
info: Adding new user `user01' (1008) with group `user01 (1008)' ...
info: Creating home directory `/home/user01' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user01
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `user01' to supplemental / extra groups `users' ...
info: Adding user `user01' to group `users' ...
dbsrv@dbsrv:~$
```

Maildir

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo -u direction mkdir /home/direction/Maildir  
dbsrv@dbsrv:~$ sudo -u secretariat mkdir /home/secretariat/Maildir  
dbsrv@dbsrv:~$ sudo -u comptabilite mkdir /home/comptabilite/Maildir  
dbsrv@dbsrv:~$ sudo -u user01 mkdir /home/user01/Maildir  
dbsrv@dbsrv:~$ sudo -u coordination mkdir /home/coordination/Maildir  
dbsrv@dbsrv:~$ sudo -u professeur mkdir /home/professeur/Maildir  
dbsrv@dbsrv:~$ sudo -u etudiants mkdir /home/etudiants/Maildir  
dbsrv@dbsrv:~$ sudo -u mediatheque mkdir /home/mediatheque/Maildir  
dbsrv@dbsrv:~$
```

Installation de Roundcube (Webmail)

Installation de composer pour gérer les dépendances de Roundcube

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo apt install composer -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  linux-headers-6.8.0-110 linux-headers-6.8.0-110-generic linux-image-6.8.0-110-generic  
  linux-modules-6.8.0-110-generic linux-modules-extra-6.8.0-110-generic linux-tools-6.8.0-110  
  linux-tools-6.8.0-110-generic  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  jsonlint php-composer-ca-bundle php-composer-class-map-generator php-composer-metadata-minifier php-composer-pcre  
  php-composer-semver php-composer-spdx-licenses php-composer-xdebug-handler php-json-schema php-psr-container  
  php-psr-log php-react-promise php-seld-signal-handler php-symfony-console php-symfony-deprecation-contracts  
  php-symfony-filesystem php-symfony-finder php-symfony-process php-symfony-service-contracts php-symfony-string unzip  
Suggested packages:  
  fossil mercurial subversion zip  
The following NEW packages will be installed:  
  composer jsonlint php-composer-ca-bundle php-composer-class-map-generator php-composer-metadata-minifier  
  php-composer-pcre php-composer-semver php-composer-spdx-licenses php-composer-xdebug-handler php-json-schema  
  php-psr-container php-psr-log php-react-promise php-seld-signal-handler php-symfony-console  
  php-symfony-deprecation-contracts php-symfony-filesystem php-symfony-finder php-symfony-process  
  php-symfony-service-contracts php-symfony-string unzip  
0 upgraded, 22 newly installed, 0 to remove and 0 not upgraded.  
Need to get 1,075 kB of archives.  
After this operation, 5,457 kB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-ca-bundle all 1.4.1-1 [11.1 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-finder all 6.4.5+dfsg-3ubuntu3 [32.0 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-pcre all 3.1.1-1 [10.2 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-class-map-generator all 1.1.0-2 [10.5 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-metadata-minifier all 1.0.0-2 [3,594 B]  
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-semver all 3.4.0-1 [19.2 kB]  
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-spdx-licenses all 1.5.0-1 [15.4 kB]  
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-psr-log all 3.0.0-1 [6,862 B]  
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-composer-xdebug-handler all 3.0.3-3 [16.6 kB]  
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-json-schema all 5.2.13-1 [33.8 kB]  
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 jsonlint all 1.10.2-1 [13.7 kB]  
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-deprecation-contracts all 3.4.1-4 [5,668 B]  
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-psr-container all 2.0.2-1 [3,878 B]  
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-service-contracts all 3.4.1-4 [10.0 kB]  
Get:15 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-string all 6.4.5+dfsg-3ubuntu3 [44.0 kB]  
Get:16 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-console all 6.4.5+dfsg-3ubuntu3 [96.7 kB]  
Get:17 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-filesystem all 6.4.5+dfsg-3ubuntu3 [30.7 kB]  
Get:18 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-symfony-process all 6.4.5+dfsg-3ubuntu3 [37.4 kB]  
Get:19 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-react-promise all 3.1.0-1 [18.5 kB]  
Get:20 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-seld-signal-handler all 2.0.2-1 [10.0 kB]  
Get:21 http://archive.ubuntu.com/ubuntu noble/universe amd64 composer all 2.7.1-2 [471 kB]  
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]  
Fetched 1,075 kB in 1s (781 kB/s)  
Selecting previously unselected package php-composer-ca-bundle.  
(Reading database ... 166587 files and directories currently installed.)  
Preparing to unpack .../00-php-composer-ca-bundle_1.4.1-1_all.deb ...  
Unpacking php-composer-ca-bundle (1.4.1-1) ...  
Selecting previously unselected package php-symfony-finder.  
Preparing to unpack .../01-php-symfony-finder_6.4.5+dfsg-3ubuntu3_all.deb ...  
Unpacking php-symfony-finder (6.4.5+dfsg-3ubuntu3) ...
```



```
Unpacking php-symfony-console (6.4.5+dfsg-3ubuntu3) ...
Selecting previously unselected package php-symfony-filesystem.
Preparing to unpack .../16-php-symfony-filesystem_6.4.5+dfsg-3ubuntu3_all.deb ...
Unpacking php-symfony-filesystem (6.4.5+dfsg-3ubuntu3) ...
Selecting previously unselected package php-symfony-process.
Preparing to unpack .../17-php-symfony-process_6.4.5+dfsg-3ubuntu3_all.deb ...
Unpacking php-symfony-process (6.4.5+dfsg-3ubuntu3) ...
Selecting previously unselected package php-react-promise.
Preparing to unpack .../18-php-react-promise_3.1.0-1_all.deb ...
Unpacking php-react-promise (3.1.0-1) ...
Selecting previously unselected package php-seld-signal-handler.
Preparing to unpack .../19-php-seld-signal-handler_2.0.2-1_all.deb ...
Unpacking php-seld-signal-handler (2.0.2-1) ...
Selecting previously unselected package composer.
Preparing to unpack .../20-composer_2.7.1-2_all.deb ...
Unpacking composer (2.7.1-2) ...
Selecting previously unselected package unzip.
Preparing to unpack .../21-unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up php-symfony-finder (6.4.5+dfsg-3ubuntu3) ...
Setting up jsonlint (1.10.2-1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Setting up php-psr-container (2.0.2-1) ...
Setting up php-composer-pcre (3.1.1-1) ...
Setting up php-symfony-service-contracts (3.4.1-4) ...
Setting up php-composer-class-map-generator (1.1.0-2) ...
Setting up php-seld-signal-handler (2.0.2-1) ...
Setting up php-composer-ca-bundle (1.4.1-1) ...
Setting up php-symfony-process (6.4.5+dfsg-3ubuntu3) ...
Setting up php-symfony-filesystem (6.4.5+dfsg-3ubuntu3) ...
Setting up php-react-promise (3.1.0-1) ...
Setting up php-psr-log (3.0.0-1) ...
Setting up php-symfony-string (6.4.5+dfsg-3ubuntu3) ...
Setting up php-composer-metadata-minifier (1.0.0-2) ...
Setting up php-composer-semver (3.4.0-1) ...
Setting up php-json-schema (5.2.13-1) ...
Setting up php-composer-spdx-licenses (1.5.8-1) ...
Setting up php-symfony-deprecation-contracts (3.4.1-4) ...
Setting up php-symfony-console (6.4.5+dfsg-3ubuntu3) ...
Setting up php-composer-xdebug-handler (3.0.3-3) ...
Setting up composer (2.7.1-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$
```

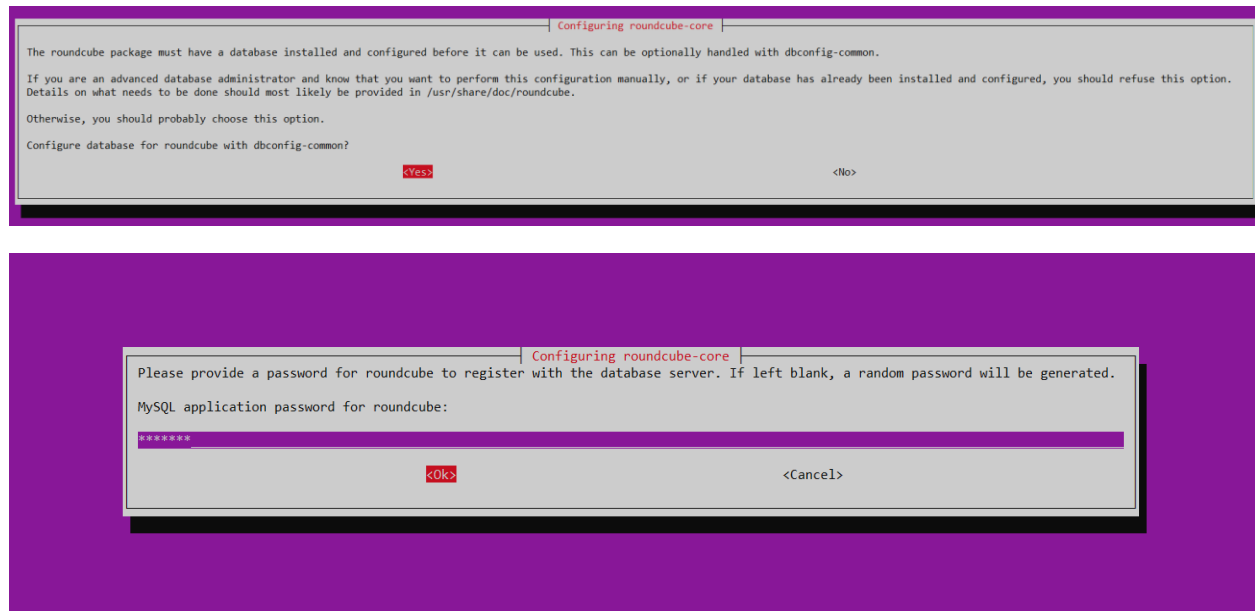
```
dbsrv@dbsrv:~$ sudo apt install apache2 php php-mysql php-intl php-mbstring php-xml unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.12).
unzip is already the newest version (6.0-28ubuntu4.1).
unzip set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-110 linux-headers-6.8.0-110-generic linux-image-6.8.0-110-generic
  linux-modules-6.8.0-110-generic linux-modules-extra-6.8.0-110-generic linux-tools-6.8.0-110-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  php php-intl php-mbstring php-mysql php-xml
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 21.0 kB of archives.
After this operation, 56.3 kB of additional disk space will be used.
Get:1 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php all 2:8.4+101~ubuntu24.04.1+deb.sury.org+1 [4,174 B]
Get:2 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php-intl all 2:8.4+101~ubuntu24.04.1+deb.sury.org+1 [4,214 B]
Get:3 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php-mbstring all 2:8.4+101~ubuntu24.04.1+deb.sury.org+1 [4,218 B]
Get:4 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php-mysql all 2:8.4+101~ubuntu24.04.1+deb.sury.org+1 [4,210 B]
Get:5 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php-xml all 2:8.4+101~ubuntu24.04.1+deb.sury.org+1 [4,224 B]
Fetched 21.0 kB in 1s (14.9 kB/s)
Selecting previously unselected package php.
(Reading database ... 167492 files and directories currently installed.)
Preparing to unpack .../php_2%3a8.4+101~ubuntu24.04.1+deb.sury.org+1_all.deb ...
Unpacking php (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Selecting previously unselected package php-intl.
Preparing to unpack .../php-intl_2%3a8.4+101~ubuntu24.04.1+deb.sury.org+1_all.deb ...
Unpacking php-intl (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Selecting previously unselected package php-mbstring.
Preparing to unpack .../php-mbstring_2%3a8.4+101~ubuntu24.04.1+deb.sury.org+1_all.deb ...
Unpacking php-mbstring (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Selecting previously unselected package php-mysql.
Preparing to unpack .../php-mysql_2%3a8.4+101~ubuntu24.04.1+deb.sury.org+1_all.deb ...
Unpacking php-mysql (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Selecting previously unselected package php-xml.
Preparing to unpack .../php-xml_2%3a8.4+101~ubuntu24.04.1+deb.sury.org+1_all.deb ...
Unpacking php-xml (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Setting up php-xml (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Setting up php (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Setting up php-mysql (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Setting up php-mbstring (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Setting up php-intl (2:8.4+101~ubuntu24.04.1+deb.sury.org+1) ...
Scanning processes...
Scanning linux images...

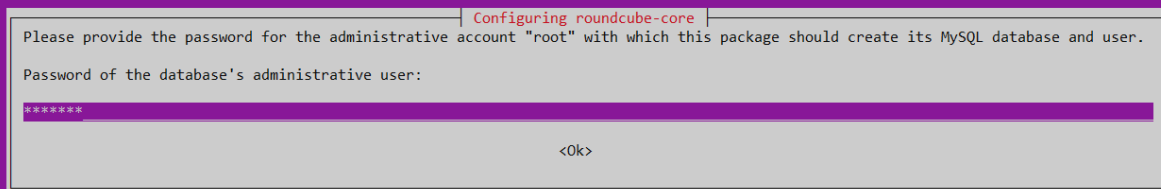
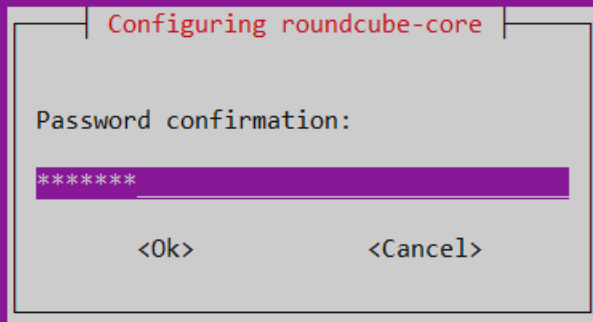
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

Figure 2-installation Apache2 et PHP





Activation du site et test de la configuration

```
dbsrv@dbsrv:~$ sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/  
dbsrv@dbsrv:~$ sudo rm /etc/nginx/sites-enabled/default  
dbsrv@dbsrv:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
dbsrv@dbsrv:~$
```

Redémarrage des services

```
dbsrv@dbsrv:~$ sudo systemctl restart php8.4-fpm  
dbsrv@dbsrv:~$ sudo systemctl restart nginx  
dbsrv@dbsrv:~$
```

Accès

← 🔒 Not secure 192.168.1.54/roundcube/



CONNEXION

Roundcube Webmail

← 🔒 Not secure 192.168.1.54/roundcube/?_task=mail&_mbox=INBOX



- Rédiger
- Courriel
- Contacts
- Paramètres
- Mode sombre
- À propos de
- Déconnexion

direction@localhost

Sélectionner Fils Options Actualiser

Boîte de réception Recherche...

La liste est vide.

<< < 1 > >>

Répondre Répondre à ... Transférer Supprimer Marquer Plus



Figure 3-Connexion avec l'utilisateur direction

Modification de /etc/hosts

```
GNU nano 7.2 /etc/hosts
192.168.1.54 mail.afj.local mail
192.168.1.54 afj.local

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Tests de fonctionnels

Depuis le terminal avec l'utilisateur : direction

```
direction@mail: ~
dbsrv@dbsrv:~$ su - direction
Password:
direction@mail:~$

direction@mail:~$ ls -la ~
total 24
drwxr-x--- 3 direction direction 4096 May 18 12:21 .
drwxr-xr-x 11 root        root      4096 May 18 12:18 ..
-rw-r--r-- 1 direction direction 220 May 18 12:05 .bash_logout
-rw-r--r-- 1 direction direction 3771 May 18 12:05 .bashrc
drwxrwxr-x 5 direction direction 4096 May 18 14:15 Maildir
-rw-r--r-- 1 direction direction 807 May 18 12:05 .profile
direction@mail:~$
```

Figure 4-le dossier Maildir existe bel et bien

```
direction@mail: ~
direction@mail:~$ ls -R ~/Maildir
/home/direction/Maildir:
cur          dovecot.index.log      dovecot-uidlist         dovecot-uidvalidity.6a0b1402  new
dovecot.index.cache  dovecot.list.index.log dovecot-uidvalidity      maildirfolder                 tmp

/home/direction/Maildir/cur:
1779113602.Vfc00I80048M491559.mail.afj.local:2,

/home/direction/Maildir/new:

/home/direction/Maildir/tmp:
direction@mail:~$
```

Figure 5-tous les dossiers sont là

```
direction@mail: ~  
direction@mail:~$ echo "Bonjour, je vous donne rendez-vous dans 30 minutes" | mail -s "Mon rendez-vous" direction  
direction@mail:~$ su - direction  
Password:  
direction@mail:~$ ls ~/Maildir/new  
1779114514.Vfc00I8004aM371790.mail.afj.local
```

Création d'un groupe mail (alias)

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo nano /etc/aliases  
dbsrv@dbsrv:~$
```

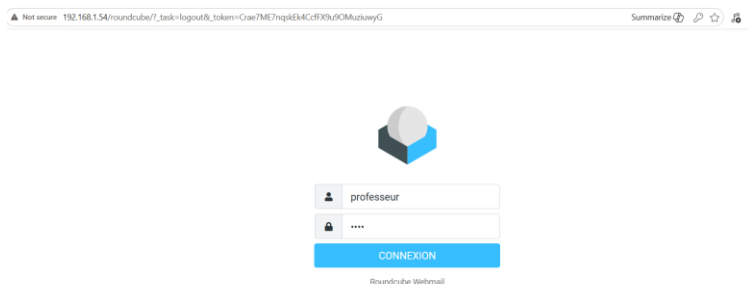
```
dbsrv@dbsrv: ~  
GNU nano 7.2 /etc/aliases *  
# See man 5 aliases for format  
itteam: direction,secretariat,comptabilite,coordination,mediatheque,etudiants,professeur,user01
```

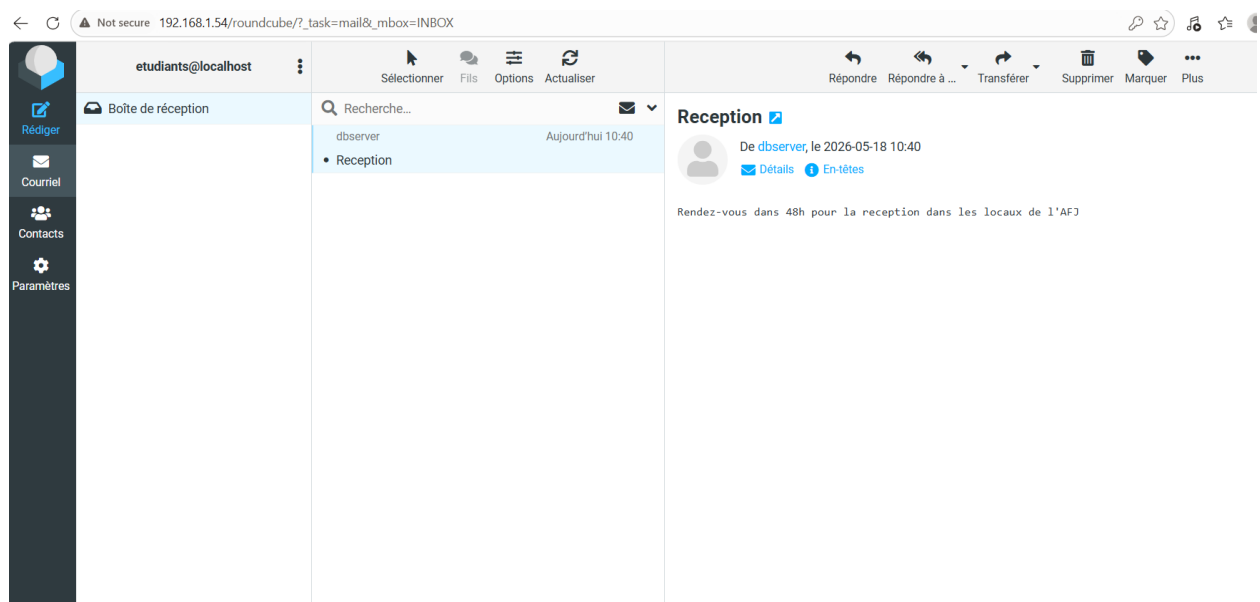
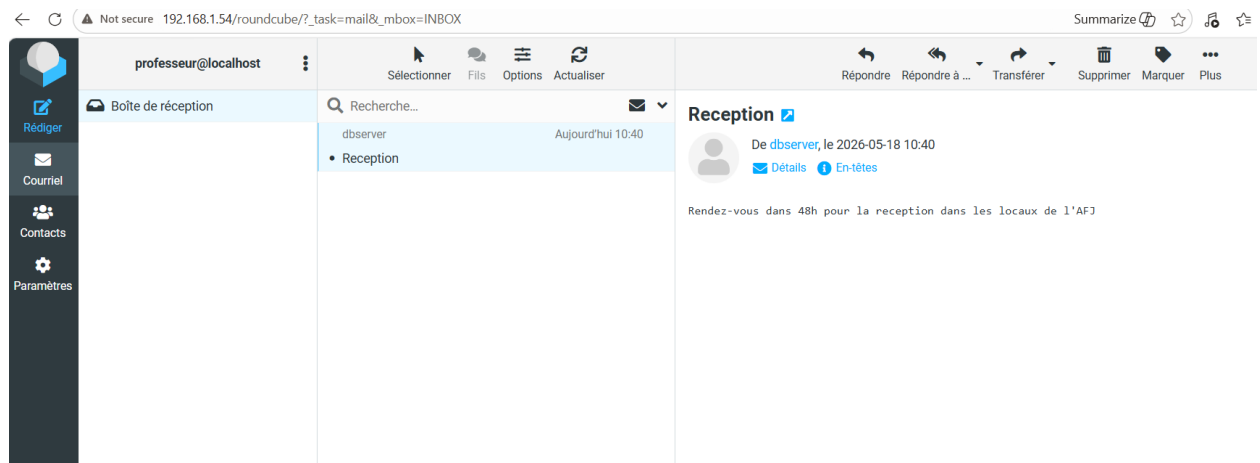
```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo newaliases  
dbsrv@dbsrv:~$
```

Envoi de message au groupe

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ echo "Rendez-vous dans 48h pour la reception dans les locaux de l'AFJ" | mail -s "Reception" itteam  
dbsrv@dbsrv:~$
```

Vérification avec Roundcube





La vérification a été faite avec deux utilisateurs : etudiants, professeur ; tout passe correctement.

Sécurisation

```
dbsrv@dbsrv: ~  
dbsrv@dbsrv:~$ sudo ufw default deny incoming  
Default incoming policy changed to 'deny'  
(be sure to update your rules accordingly)  
dbsrv@dbsrv:~$ sudo ufw default allow outgoing  
Default outgoing policy changed to 'allow'  
(be sure to update your rules accordingly)  
dbsrv@dbsrv:~$ sudo ufw allow 22  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 80  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 8080  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 443  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 25/tcp  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 587/tcp  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 993/tcp  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 465  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 110  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 995  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$ sudo ufw allow 143  
Rules updated  
Rules updated (v6)  
dbsrv@dbsrv:~$
```

```
dbsrv@dbsrv:~$ sudo ufw enable  
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y  
Firewall is active and enabled on system startup  
dbsrv@dbsrv:~$ sudo ufw status verbose  
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), disabled (routed)  
New profiles: skip  
  
To Action From  
--  
22/tcp ALLOW IN Anywhere  
22 ALLOW IN Anywhere  
80 ALLOW IN Anywhere  
8080 ALLOW IN Anywhere  
443 ALLOW IN Anywhere  
25/tcp ALLOW IN Anywhere  
587/tcp ALLOW IN Anywhere  
993/tcp ALLOW IN Anywhere  
465 ALLOW IN Anywhere  
110 ALLOW IN Anywhere  
995 ALLOW IN Anywhere  
143 ALLOW IN Anywhere  
22/tcp (v6) ALLOW IN Anywhere (v6)  
22 (v6) ALLOW IN Anywhere (v6)  
80 (v6) ALLOW IN Anywhere (v6)  
8080 (v6) ALLOW IN Anywhere (v6)  
443 (v6) ALLOW IN Anywhere (v6)  
25/tcp (v6) ALLOW IN Anywhere (v6)  
587/tcp (v6) ALLOW IN Anywhere (v6)  
993/tcp (v6) ALLOW IN Anywhere (v6)  
465 (v6) ALLOW IN Anywhere (v6)  
110 (v6) ALLOW IN Anywhere (v6)  
995 (v6) ALLOW IN Anywhere (v6)  
143 (v6) ALLOW IN Anywhere (v6)  
dbsrv@dbsrv:~$
```


Analyse des logs

```
dbsrv@dbsrv:~$ sudo tail -f /var/log/mail.log
2026-05-18T14:46:35.898705+00:00 dbsrv dovecot: imap-login: Login: user=<etudiants>, method=PLAIN, rip=::1, lip=::1, mpid=10973, secured, session=<BSYboRhSfKQA
AAAAAAAAAAAAAAAAAAB>
2026-05-18T14:46:35.901588+00:00 dbsrv dovecot: imap(etudiants)<10973><BSYboRhSfKQAAAAAAAAAAAAAAAAAAB>: Disconnected: Logged out in=90 out=971 deleted=0 expu
nged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
2026-05-18T14:48:42.805907+00:00 dbsrv dovecot: imap-login: Login: user=<etudiants>, method=PLAIN, rip=::1, lip=::1, mpid=10977, secured, session=<Q0erqBhSzuMA
AAAAAAAAAAAAAAAAAAB>
2026-05-18T14:48:42.810114+00:00 dbsrv dovecot: imap(etudiants)<10977><Q0erqBhSzuMAAAAAAAAAAAAAAAAAAAB>: Disconnected: Logged out in=90 out=971 deleted=0 expu
nged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
2026-05-18T14:49:35.899429+00:00 dbsrv dovecot: imap-login: Login: user=<etudiants>, method=PLAIN, rip=::1, lip=::1, mpid=10979, secured, session=<Ma/VqxhS6qsA
AAAAAAAAAAAAAAAAAAB>
2026-05-18T14:49:35.905713+00:00 dbsrv dovecot: imap(etudiants)<10979><Ma/VqxhS6qsAAAAAAAAAAAAAAAAAAB>: Disconnected: Logged out in=90 out=971 deleted=0 expu
nged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
2026-05-18T14:50:35.894365+00:00 dbsrv dovecot: imap-login: Login: user=<etudiants>, method=PLAIN, rip=::1, lip=::1, mpid=11027, secured, session=<6zJprxhS+NsA
AAAAAAAAAAAAAAAAAAB>
2026-05-18T14:50:35.898994+00:00 dbsrv dovecot: imap(etudiants)<11027><6zJprxhS+NsAAAAAAAAAAAAAAAAAAB>: Disconnected: Logged out in=90 out=971 deleted=0 expu
nged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
2026-05-18T14:51:36.912539+00:00 dbsrv dovecot: imap-login: Login: user=<etudiants>, method=PLAIN, rip=::1, lip=::1, mpid=11034, secured, session=<ch/0LsxhStu0A
AAAAAAAAAAAAAAAAAAB>
2026-05-18T14:51:36.914996+00:00 dbsrv dovecot: imap(etudiants)<11034><ch/0LsxhStu0AAAAAAAAAAAAAAAAAAB>: Disconnected: Logged out in=90 out=971 deleted=0 expu
nged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
```

Pour plus de sécurité

Installation de fail2ban

```
dbsrv@dbsrv:~$ sudo apt install fail2ban
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-110 linux-headers-6.8.0-110-generic linux-image-6.8.0-110-generic linux-modules-6.8.0-110-generic linux-modules-extra-6.8.0-110-generic
  linux-tools-6.8.0-110 linux-tools-6.8.0-110-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 496 kB of archives.
After this operation, 2,572 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyasyncore all 1.0.2-2 [10.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 fail2ban all 1.0.2-3ubuntu0.1 [409 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyinotify all 0.9.6-2ubuntu1 [25.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd64 5.5.22 [51.7 kB]
Fetched 496 kB in 0s (1,369 kB/s)
Selecting previously unselected package python3-pyasyncore.
(Reading database ... 174001 files and directories currently installed.)
Preparing to unpack .../python3-pyasyncore_1.0.2-2_all.deb ...
Unpacking python3-pyasyncore (1.0.2-2) ...
Selecting previously unselected package fail2ban.
Preparing to unpack .../fail2ban_1.0.2-3ubuntu0.1_all.deb ...
Unpacking fail2ban (1.0.2-3ubuntu0.1) ...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack .../python3-pyinotify_0.9.6-2ubuntu1_all.deb ...
Unpacking python3-pyinotify (0.9.6-2ubuntu1) ...
Selecting previously unselected package whois.
Preparing to unpack .../whois_5.5.22_amd64.deb ...
Unpacking whois (5.5.22) ...
Setting up whois (5.5.22) ...
Setting up python3-pyasyncore (1.0.2-2) ...
Setting up fail2ban (1.0.2-3ubuntu0.1) ...
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:224: SyntaxWarning: invalid escape sequence '\s'
  "1490349000 test failed.dns.ch", "\s*test <F-ID>\S+</F-ID>"
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:435: SyntaxWarning: invalid escape sequence '\S'
```

```

Setting up whois (5.5.22) ...
Setting up python3-pyasyncore (1.0.2-2) ...
Setting up fail2ban (1.0.2-3ubuntu0.1) ...
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:224: SyntaxWarning: invalid escape sequence '\s'
  "1490349000 test failed.dns.ch", "^\\s*test <F-ID>\S+</F-ID>"
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:435: SyntaxWarning: invalid escape sequence '\S'
  '^'+prefix+'<F-ID>User <F-USER>\S+</F-USER></F-ID> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:443: SyntaxWarning: invalid escape sequence '\S'
  '^'+prefix+'User <F-USER>\S+</F-USER> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:444: SyntaxWarning: invalid escape sequence '\d'
  '^'+prefix+'Received disconnect from <F-ID><ADDR> port \d+</F-ID>'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:451: SyntaxWarning: invalid escape sequence '\s'
  _test_variants('common', prefix="\s*\S+ sshd\[<F-MLFID>\d+</F-MLFID>\]:\s+")
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregex testcase.py:537: SyntaxWarning: invalid escape sequence '\['
  'common[prefix="^svc\[<F-MLFID>\d+</F-MLFID>\] connect <F-CONTENT>.+</F-CONTENT>$"'
/usr/lib/python3/dist-packages/fail2ban/tests/server testcase.py:1375: SyntaxWarning: invalid escape sequence '\s'
  "{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)$'; } | while read -r hdl; do\"",
/usr/lib/python3/dist-packages/fail2ban/tests/server testcase.py:1378: SyntaxWarning: invalid escape sequence '\s'
  "{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)$'; } | while read -r hdl; do\"",
/usr/lib/python3/dist-packages/fail2ban/tests/server testcase.py:1421: SyntaxWarning: invalid escape sequence '\s'
  "{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)$'; } | while read -r hdl; do\"",
/usr/lib/python3/dist-packages/fail2ban/tests/server testcase.py:1424: SyntaxWarning: invalid escape sequence '\s'
  "{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)$'; } | while read -r hdl; do\"",
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
Setting up python3-pyinotify (0.9.6-2ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
dbsrv@dbsrv:~$ sudo nano /etc/fail2ban/jail.local
dbsrv@dbsrv:~$

```

```

dbsrv@dbsrv: ~
GNU nano 7.2 /etc/fail2ban/jail.local *
[postfix]
enabled = true
port = smtp,smtps,submission
filter = postfix
logpath = /var/log/mail.log
maxretry = 5
bantime = 3600

[dovecot]
enabled = true
port = pop3,pop3s,imap,imaps
filter = dovecot
logpath = /var/log/mail.log
maxretry = 5
bantime = 3600

```

```

dbsrv@dbsrv: ~
dbsrv@dbsrv:~$ sudo systemctl restart fail2ban
dbsrv@dbsrv:~$ sudo fail2ban-client status
Status
|- Number of jail:      3
|- Jail list:   dovecot, postfix, sshd
dbsrv@dbsrv:~$

```

Utilisation de ufw et fail2ban

Ces deux outils sont utilisés pour renforcer la sécurité sur mon serveur. Ufw pour bloquer les ports non autorisés en limitant l'accessibilité. Fail2ban de son côté surveille les logs et bannit tout comportement suspect ou tout échec de connexion répétitif.